

STAT 2020 Discussion Slides

Week 15: Two-Sample Proportion Test and Chi-Squared Test for Independence

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SPOT Survey Reminder

Please remind students:

- If more than 50% of the class completes SPOT, everyone gets an additional 5% increase on the quiz portion (about 1 point).
- This is on top of the previously promised 5% quiz curve.
- Encourage students to complete it as soon as possible.

Agenda

- Brief announcement and logistics
- Two-sample proportion hypothesis test (for interest; not on final)
- Review chi-squared test for independence with full 3-step workflow
- Work one complete 3×2 chi-squared example
- If extra time: categorical vs quantitative data; chi-squared distribution shape
- Quick attendance check

Two-Sample Proportion Test: Context

Compare proportions from two independent groups:

$$p_1 - p_2.$$

Typical hypotheses:

$$H_0 : p_1 - p_2 = 0 \quad \text{vs.} \quad H_a : p_1 - p_2 > 0, \quad p_1 - p_2 < 0, \quad \text{or} \quad p_1 - p_2 \neq 0.$$

Course note: this topic is included for student interest and will **not** be on the final exam.

Score Test Formula (Pooled Under H_0)

Let

$$\hat{p}_1 = \frac{x_1}{n_1}, \quad \hat{p}_2 = \frac{x_2}{n_2}, \quad \tilde{p} = \frac{x_1 + x_2}{n_1 + n_2}.$$

Under $H_0 : p_1 = p_2$, use pooled SE:

$$Z^* = \frac{(\hat{p}_1 - \hat{p}_2) - 0}{\sqrt{\tilde{p}(1 - \tilde{p}) \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}}.$$

Right-tailed decision:

$$\text{Reject } H_0 \text{ if } Z^* > z_\alpha, \quad p\text{-value} = P(Z > Z^*).$$

Conditions Checklist

- Two random samples (or random assignment in each group)
- Independence within each group and between groups
- Normal approximation is reasonable under H_0 :

$n_1\tilde{p}$, $n_1(1 - \tilde{p})$, $n_2\tilde{p}$, $n_2(1 - \tilde{p})$ are all sufficiently large (rule of thumb: ≥ 5).

Worked Example

Example: Two-Sample Proportion (Right-Tailed)

New tutoring app vs. standard method:

$$x_1 = 54, n_1 = 90, \hat{p}_1 = 0.60; \quad x_2 = 36, n_2 = 90, \hat{p}_2 = 0.40.$$

Test at $\alpha = 0.05$:

$$H_0 : p_1 - p_2 = 0 \quad \text{vs.} \quad H_a : p_1 - p_2 > 0.$$

Step 2:

$$\tilde{p} = \frac{54 + 36}{180} = 0.50, \quad Z^* = \frac{0.60 - 0.40}{\sqrt{0.5(0.5)(1/90 + 1/90)}} = 2.68.$$

$$z_{0.05} = 1.645, \quad 2.68 > 1.645 \Rightarrow \text{reject } H_0.$$

Step 3: evidence supports a higher success proportion with the new app.

p-value = $P(Z > 2.68) \approx 0.0037$.

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CI for $p_1 - p_2$ (Unpooled SE)

For a two-sided $(1 - \alpha)100\%$ CI:

$$(\hat{p}_1 - \hat{p}_2) \pm z_{\alpha/2} \sqrt{\frac{\hat{p}_1(1 - \hat{p}_1)}{n_1} + \frac{\hat{p}_2(1 - \hat{p}_2)}{n_2}}.$$

For the tutoring example:

$$0.20 \pm 1.96 \sqrt{\frac{0.6 \cdot 0.4}{90} + \frac{0.4 \cdot 0.6}{90}} = 0.20 \pm 0.143.$$

95% CI for $p_1 - p_2$: (0.057, 0.343).

3-Step Template

Chi-Squared Independence Test: 3-Step Template

Step 1 (Setup):

 H_0 : Row variable and column variable are independent H_a : They are not independent.

Symbol	Meaning
o_{ij}	observed count in cell (i, j)
e_{ij}	expected count in cell (i, j)
r, c	number of rows, columns

Step 2 (Decision Rule):

$$e_{ij} = \frac{(\text{row total})(\text{column total})}{n}, \quad \chi^{2*} = \sum \frac{(o_{ij} - e_{ij})^2}{e_{ij}}.$$

$$\text{df} = (r - 1)(c - 1), \quad \text{Reject } H_0 \text{ if } \chi^{2*} > \chi^2_{\alpha, \text{df}}.$$

Step 3 (Conclude): compare by critical value or by p-value.

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Worked 3x2 Example

Example Data (3x2 Table)

Study style vs. pass/fail on a quiz:

Study Style	Pass	Fail	Row Total
Notes only	30	10	40
Videos only	18	22	40
Both methods	27	13	40
Column Total	75	45	120

Under H_0 , expected counts for each row are:

$$E(\text{Pass}) = 40 \cdot \frac{75}{120} = 25, \quad E(\text{Fail}) = 40 \cdot \frac{45}{120} = 15.$$

Example (Cont.): Test Statistic and Decision

Compute

$$\begin{aligned}\chi^{2*} = \sum \frac{(o - e)^2}{e} &= \frac{(30 - 25)^2}{25} + \frac{(10 - 15)^2}{15} \\ &+ \frac{(18 - 25)^2}{25} + \frac{(22 - 15)^2}{15} + \frac{(27 - 25)^2}{25} + \frac{(13 - 15)^2}{15} = 8.32.\end{aligned}$$

$$df = (3 - 1)(2 - 1) = 2, \quad \chi_{0.05, 2}^2 = 5.99.$$

Since $8.32 > 5.99$, reject H_0 .

p-value = $P(\chi_2^2 > 8.32) \approx 0.016$.

Conclusion: study style and quiz outcome are associated.

Categorical vs. Quantitative Data (Quick Review)

Categorical:

- Labels/groups (major, class level, treatment group)
- Arithmetic differences are not meaningful

Quantitative:

- Numeric measurements (height, score, waiting time)
- Differences and averages are meaningful

Chi-Squared Distribution: Intuition

- χ^2 is always nonnegative and right-skewed.
- As df increases, the curve shifts right and becomes less skewed.
- Common $\alpha = 0.05$ critical values:

$$\chi_{0.05,1}^2 = 3.84, \quad \chi_{0.05,2}^2 = 5.99, \quad \chi_{0.05,3}^2 = 7.81, \quad \chi_{0.05,5}^2 = 11.07.$$

Final Exam Coverage (If Asked)

After chi-squared test, remaining topics are:

- Inference for σ^2
- Linear regression

These are included in final exam coverage.

Take-Home Messages

Take-Home Messages

- Two-sample proportion test uses pooled SE under H_0 .
- For chi-squared independence tests, always compute expected counts first.
- Keep the 3-step framework: setup, decision rule, compute and conclude.
- For this week, focus on clean setup and correct test choice.

In-Class Attendance Check (Very Quick)

In a chi-squared independence test, suppose:

$$df = 2, \quad \alpha = 0.05, \quad \chi^{2*} = 6.2, \quad \chi^2_{0.05,2} = 5.99.$$

Decision?

Please submit: Full Name; Section Number; Your Answer

In-Class Attendance Check (Very Quick)

In a chi-squared independence test, suppose:

$$df = 2, \quad \alpha = 0.05, \quad \chi^{2*} = 6.2, \quad \chi^2_{0.05,2} = 5.99.$$

Decision?

Please submit: Full Name; Section Number; Your Answer

Answer (1–2 words): Reject H_0 .